Dr. Juliane Hennecke

Chair of Empirical Microeconomics

MARTIN-LUTHER-UNIVERSITÄT
HALLE-WITTENBERG

ADVANCED MICROECONOMICS

Decisions under Risk and Uncertainty III - Implicit Market for Risk and Value of a Statistical Life

Winter Term 2025/2026

Outline

Literature

- Autor, David (2016): Lecture Notes 18
- Ashenfelter, O., & Greenstone, M. (2004): Using Mandated Speed Limits to Measure the Value of a Statistical Life. Journal of Political Economy, 112(S1), Pages 226–263

Overview

Implicit Market for Risk

From Expected Utility to Real-World Risk Trade-offs

- **So far:**
  - Expected utility theory: How individuals evaluate risky lotteries.
  - Risk preferences: How utility curvature determines willingness to bear risk.
  - Insurance markets: Explicit markets where risk is traded.
- **Today:**
  - Many risks are not traded via insurance contracts.
  - Instead, risk is traded **implicitly** through everyday decisions.
- Goal of this lecture:
  - Understand how these choices reveal the **value of safety**.

Implicit Market for Risk: Concept

- Not all risk trading occurs in formal insurance markets.
- In many settings, individuals face trade-offs between:
  - money, time, or convenience
  - and the probability of loss (e.g. financial loss, damage or injury/death)

→ These trade-offs generate **implicit prices of risk**.

- Observing choices allows us to infer:
  - willingness to pay (WTP) for risk reduction,
  - willingness to accept (WTA) higher risk.

Examples of Implicit Risk Markets

- **Consumption choices**:
  - Buying safety products (helmets, child car seats, airbags)
  - Choosing safer vs. riskier transportation modes
- **Labor market choices**:
  - Accepting higher wages in exchange for higher injury/fatality risk
- **Policy choices**:
  - Speed limits,
  - Safety regulations.
- In all cases: Higher safety/lower risk $\rightarrow$ Higher cost
- **Revealed preferences:** Choices reveal how much cost an agent accepts to reduce risk.

Overview

Safety Products as Implicit Risk Markets

- Many products reduce the **probability or severity of harm**.
- **Examples:**
  - child car seats,
  - bicycle helmets,
  - smoke detectors,
  - home security systems.
- No explicit market for “risk reduction” exists.
- Instead: Consumers **buy safety by paying higher prices**.
- Purchases reveal willingness to pay for small reductions in risk.
- **Policy implication:** Aggregating across consumers allows economists to infer the **average willingness to pay for a (statistical) life saved**

Overview

Risk in Labor Markets

- Jobs differ in **working conditions** and risk levels.
- Some jobs involve lower/higher probability of injury or death:

Table 1
Annual occupational fatality risk per 1000 fulltime-equivalent-man-years of the 10 most dangerous occupations, 1985–1995.

Rank	Occupation (3-digit)	Mean	Std.Dev.	Min.	Max.
1	Scaffolders	0.798	0.364	0.186	1.56
2	Inland waters navigator/sundry waterways occupations	0.714	0.249	0.348	1.10
3	Deckhands	0.681	0.428	0.135	1.37
4	Nautical navigators	0.513	0.334	0.155	1.07
5	Roofers, slaters	0.418	0.128	0.179	0.610
6	Miners	0.361	0.132	0.146	0.653
7	Machine, electrical and shot colliers	0.331	0.254	0.000	0.945
8	Air traffic occupations	0.290	0.225	0.000	0.732
9	Blasters/sundry civil engineering occupations	0.277	0.069	0.125	0.404
10	Excavator drivers	0.267	0.097	0.110	0.406

Source: Schaffner, S. and H. Spengler (2010). Using job changes to evaluate the bias of value of a statistical life estimates. Resource and Energy Economics 32(1): 15–27.

- In competitive labor markets riskier jobs must offer higher wages.
- This wage premium is called a **compensating wage differential**, as it compensates for the higher risk (or any other non-monetary job characteristic).

Compensating Differentials and Risk Preferences

- (More) Risk-averse workers require (higher) compensation for risk.
- The wage–risk trade-off reflects the **marginal rate of substitution between income and mortality risk**.
- Less (more) risk-averse workers, and firms for which productivity in risky occupations is relatively high (low), form **efficient matches**.
- **Policy implication:** Aggregating across workers allows economists to infer the **average willingness to pay for a (statistical) life saved**

Overview

Safety Products and Public Policy

- Policymakers often consider:
  - mandating safety (products),
  - subsidizing them,
  - or providing information.
- **Theoretical prediction:** Mandating more safety is **not necessarily welfare-improving** if consumers have **full information** and no externalities are present.
  -
  - Consumers (workers) who are fully informed about risks choose the optimal level of risk based on their preferences / valuation of safety → "Forcing" them to choose differently reduces welfare.
  - Behavioral responses matter: Individuals with a low valuation of safety (risk loving) may substitute toward other, riskier products/occupations.

Why Safety Regulation?

- In standard theory: Regulations restrict choice $\rightarrow$ potential welfare loss
- **Market failure:** Implicit risk markets may fail.
- Reasons:
  - imperfect information about risks,
  - behavioral biases (underestimation of low-probability risks),
  - externalities across workers or consumers.
- **Result:** Privately chosen risk levels may be inefficient.
- If risks are misperceived, regulation can **increase welfare**
- **Safety standards** typically reduce risk but increase costs for firms and/or workers.

Example: Child Car Seats on Airplanes

- U.S. National Transportation Safety Board (NTSB) proposal: Require infants <2 years to sit in child safety restraints (CSR) on airplanes (no free travel on adult's lap)
- Theoretical prediction: If parents have full information about the risks of traveling with their babies on their laps, parents will make the utility maximizing decision (between saving money and safety of their babies).
- **Market failure** possible - Incomplete information (about risks)
- Regulation might push parents to privately optimal decisions conditional on air travel
- But: Regulation increases travel costs $\rightarrow$ some families switch to car travel.
- Car risk **far higher** $\rightarrow$ net effect may increase total fatalities.

Marginal Value of Public Funds

Marginal Value of Public Funds

MVPF calculates a **policy's social benefit** per dollar of net government cost, i.e. the amount of money individuals are **willing to pay** (WTP) for the policy change relative to its costs.

$$\text{MVPF} = \frac{\text{WTP}}{\text{Government Costs}}$$

- The MVPF compares individuals' willingness to pay to the fiscal cost of a policy.
- In addition to the **WTP** the **government costs** are needed to calculate the MVPF.
- Two types of costs:
  1. Actual (direct) costs of the policy change
  2. Indirect costs due to behavioral changes

Overview

The Value of a Statistical Life (VSL)

- Revealed choices (of consumers and workers) can be used to calculate the willingness to pay for safety (e.g. for MVPF) using the concept of the **value of a statistical life**

Value of a statistical life

The monetary value that individuals implicitly place on reducing mortality risk.

- VSL is **not** the value of a specific human life; it is inferred from risk–money tradeoffs.
- **Statistical life:** An anonymous, probabilistic life—not a specific individual.

Revealed Preference in Safety Decisions

- Individuals or governments reveal how they value safety by the trade-offs they choose.
- We observe VSL through:
  - wage-risk tradeoffs (occupational safety regulations)
  - product choices
  - regulatory decisions (e.g. environmental regulations)
  - transportation behavior (speed limits)
- States raising speed limits implicitly reveal:

$$\text{value of time saved} \geq \text{value of statistical lives lost}$$

- We can measure VSL using these **revealed choices**.

Overview

Ashenfelter, O., & Greenstone, M. (2004)

Ashenfelter, O., & Greenstone, M. (2004): Using Mandated Speed Limits to Measure the Value of a Statistical Life. Journal of Political Economy, 112(S1), Pages 226–263

Using Mandated Speed Limits to Measure the Value of a Statistical Life

---

Orley Ashenfelter

Princeton University and National Bureau of Economic Research

Michael Greenstone

Massachusetts Institute of Technology, National Bureau of Economic Research, and American Bar Foundation

Ashenfelter, O., & Greenstone, M. (2004)

- We will discuss the following questions:
  - What is the paper about?
  - How is it related to our class?
  - What data and methods are used?
  - What are the paper's (main) results?
  - How valid are the paper's findings?
  - What is your personal take-away from this paper?

Summary

- Many risks are not traded in explicit insurance markets but through **implicit markets**.
- Individuals trade off money, time, or convenience against risk in everyday decisions.
- These trade-offs reveal the **willingness to pay for safety**.
- Labor markets feature **compensating wage differentials** for risky jobs.
- Aggregating revealed preferences allows economists to estimate the **Value of a Statistical Life (VSL)**.
- Public safety regulation may increase welfare when implicit risk markets fail due to
  - imperfect information,
  - behavioral biases,
  - or externalities.

Keywords

- Expected Utility Theory
- Implicit Market for Risk
- Willingness to Pay (WTP)
- Willingness to Accept (WTA)
- Safety Products
- Compensating Wage Differentials
- Risk Preferences
- Market Failure
- Value of a Statistical Life (VSL)
- Revealed Preferences
- Marginal Value of Public Funds (MVPF)

Questions?